

Artful Computing Workshop – Additional Tool Installation

For the first half of the workshop, all you need is **NetLogo 7.x** and a creative mindset.

BUT for the **SECOND HALF**, if you want to make more advanced 3D artistic renderings (with shadows and reflections), and make movies from those, you must install third party software.

For LINUX machines:

1. Install POV-Ray **sudo apt install povray**
(for Debian-based distros. For other distros, check your package manager)
2. Install OpenShot GUI video editor from <https://www.openshot.org/>
OR **ffmpeg** command line tool for video creation: **sudo apt install ffmpeg**

For WINDOWS machines:

1. Download POV-Ray 3.7 installer (<https://www.povray.org/download/>), run it,
and also run the extra "editor" installer the first installer will prompt you to download!
2. Install OpenShot video editor from <https://www.openshot.org/>
OR download **ffmpeg.exe** from <https://tinyurl.com/ffmpegexe> and put it in the same folder as the other workshop materials.

For MAC machines:

On Mac, installing **POV-Ray 3.7** requires a package manager like **homebrew** or **macports**. Homebrew is more popular in general, so let's try that.

1. If you don't have it already, install the "Homebrew" package manager for Mac.
<https://brew.sh/>
(You might have to install Xcode command line tools first: **xcode-select -install**)
2. From the terminal, run: **brew install povray**
3. Install OpenShot video editor from <https://www.openshot.org/>
OR **brew install ffmpeg** (if you prefer command line video creation vs. OpenShot)